IFRS 9 Expected Credit Loss Pipeline

PD $\times$ LGD $\times$ EAD across Macro Scenarios

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1 What this project does

Expected Credit Loss (ECL) under IFRS 9 is the accounting provision for future credit losses on a lending portfolio. It combines the three components estimated in the sibling projects into a single provision number:

$$\mathrm{ECL} = \sum_{s \in \mathrm{scenarios}} w_s \cdot \mathrm{PD}_s \cdot \mathrm{LGD}_s \cdot \mathrm{EAD} \cdot \mathrm{DiscountFactor}.$$

This project runs the full IFRS 9 pipeline on a synthetic retail portfolio of 50 000 performing loans with total exposure of 3.78 billion EUR across five products (mortgage, auto, personal loan, credit card, overdraft). It applies stage classification, applies three probability-weighted macro scenarios, and produces the provision number that would post to the balance sheet.

2 Data

2.1 Portfolio

Product	Loans	Mean PD 12m	Mean LGD	Total EAD (m EUR)
Mortgage	17 411	2.6%	22%	3 399.5
Auto	7 581	4.9%	43%	167.7
Personal loan	10 062	8.2%	71%	112.1
Credit card	9 865	9.6%	71%	71.6
Overdraft	5 081	12.3%	71%	29.7

Table 1: Portfolio characteristics by product. Mortgages dominate EAD by an order of magnitude despite lower default rates; unsecured revolving products carry high PD and LGD but small balances.

2.2 Macro scenarios

Scenario	Weight	PD mult.	LGD mult.	Unemp %	HPI YoY %
Baseline	60%	1.00	1.00	7.5	+2.5
Adverse	30%	1.60	1.20	9.0	-2.0
Severe adverse	10%	2.40	1.50	11.5	-6.0

Table 2: Three probability-weighted macro paths. Multipliers are applied to per-loan PD and LGD before ECL calculation, then scenario-level ECLs are combined by weight to produce the final IFRS 9 provision.

3 Methodology

3.1 IFRS 9 staging

- **Stage 1:** performing loans with no significant deterioration since origination. ECL is 12-month PD $\times$ LGD $\times$ EAD, discounted one year.
- **Stage 2:** significant deterioration (credit-score drop $\geq$ 60 points OR watchlist flag) OR DPD between 30 and 89 days. ECL is lifetime.

- **Stage 3:** DPD $\geq$ 90 days (credit-impaired). PD set to 1; ECL is lifetime LGD $\times$ EAD.

The performing portfolio in this project shows 76.3% in stage 1, 23.0% in stage 2, and 0.8% in stage 3.

3.2 Lifetime PD

Lifetime PD across remaining maturity T (5 years for term loans, 3 years for revolving) is computed under a constant-hazard assumption:

$$\mathrm{PD}_{\mathrm{lifetime}} = 1 - (1 - \mathrm{PD}_{12m})^T.$$

3.3 Discounting

Cash flows are discounted at an effective annual rate of 3%. For lifetime ECL the discount factor is the average of annual factors across the remaining maturity.

4 Results

4.1 Portfolio ECL under three scenarios

Scenario	ECL (m EUR)	Coverage vs EAD
Baseline	77.5	2.05%
Adverse	132.5	3.51%
Severe adverse	217.9	5.77%
Weighted (final IFRS 9 provision)	**108.0**	**2.86%**

Table 3: Portfolio ECL across the three macro scenarios and the probability-weighted final provision. The adverse scenario adds +71% over baseline; severe adverse adds +181%. Weighting brings the final provision to 108m EUR, dominated by the 60% baseline contribution but lifted by the tail scenarios.

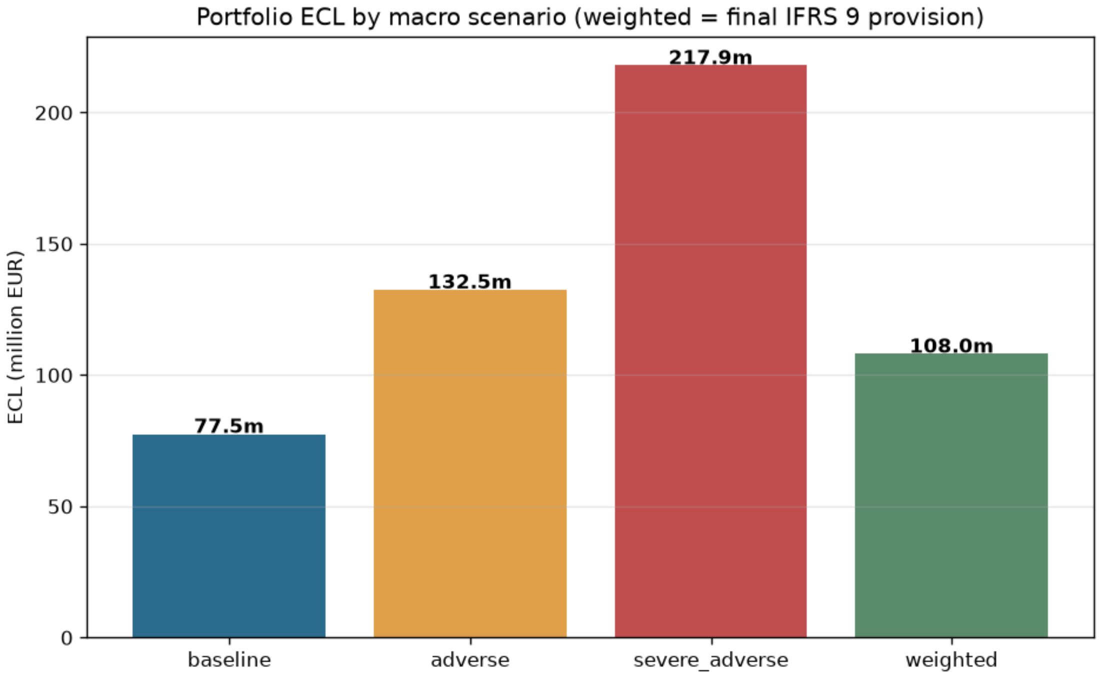

Figure 1: Portfolio ECL under each scenario and the weighted final provision. The weighted number is what gets posted to the balance sheet as impairment allowance.

4.2 ECL by IFRS 9 stage

Stage	Loans	Total EAD (m)	Weighted ECL (m)	Coverage
1	38 140	2 881.7	31.7	1.10%
2	11 483	868.2	67.8	7.81%
3	377	30.7	8.6	28.05%

Table 4: ECL split by IFRS 9 stage. Stage 2 alone carries 63% of the total weighted provision despite representing only 23% of loan count and 23% of EAD, reflecting the lifetime-versus-12-month difference. Stage 3 has the highest coverage ratio because PD = 1 by construction.

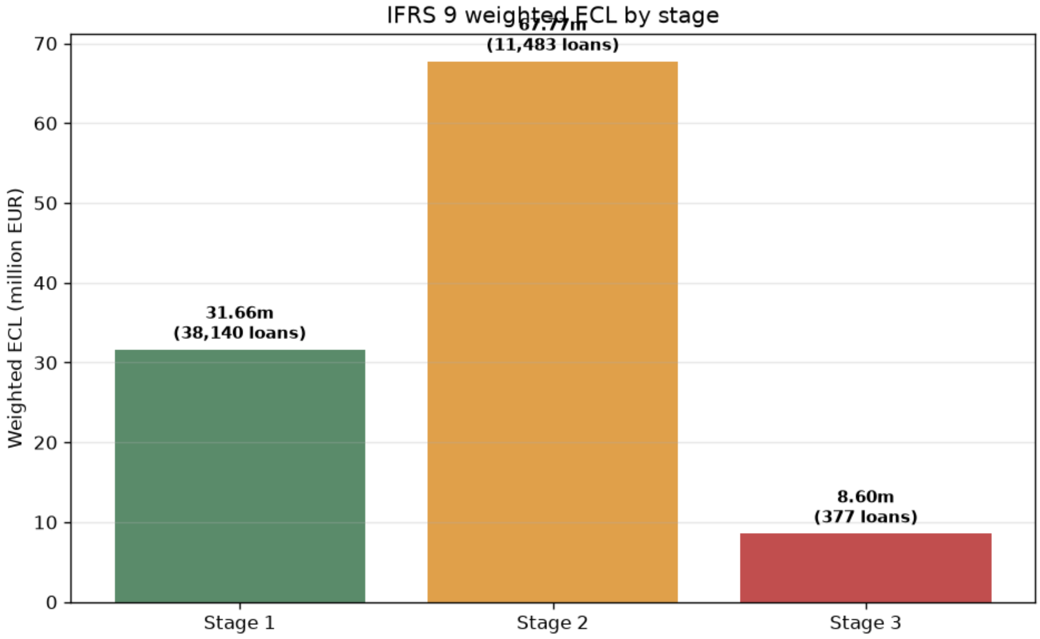

Figure 2: Weighted ECL by IFRS 9 stage. The middle stage carries the largest absolute provision, which is the standard shape for IFRS 9 retail books because the lifetime multiplier on non-defaulted-but- deteriorated exposures compounds a lot of provisioning weight.

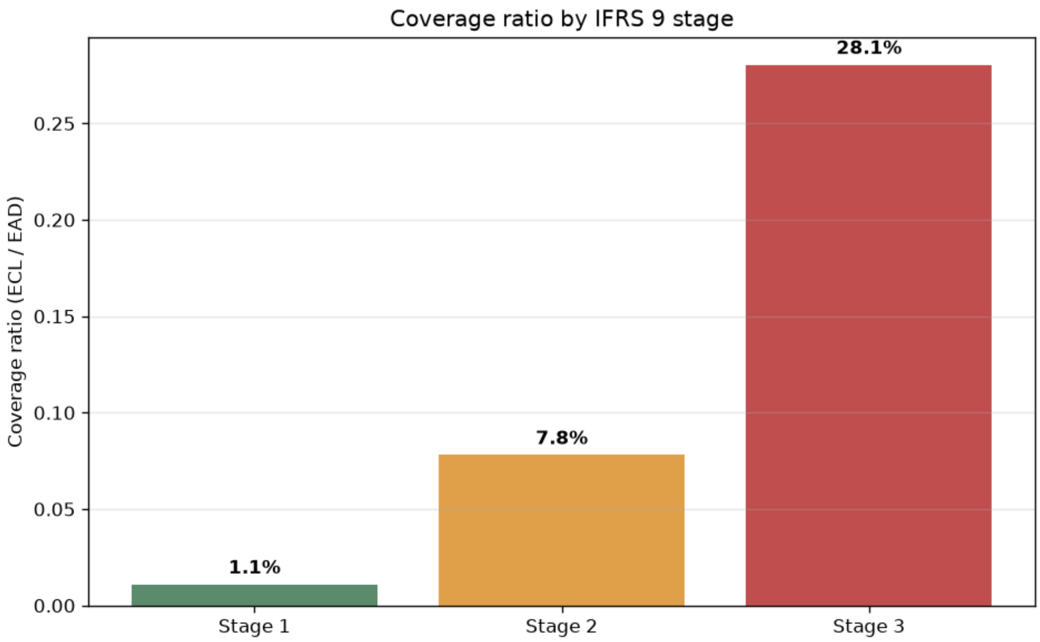

Figure 3: Coverage ratio (weighted ECL / EAD) by stage. The approximately 8x step from stage 1 to stage 2, and the further 3.6x step to stage 3, quantifies the "cliff effect" that IFRS 9 policy teams monitor closely.

4.3 ECL by product

Product	Total EAD (m)	Weighted ECL (m)	Coverage	PD × LGD baseline
Mortgage	3 399.5	66.2	1.95%	0.57%
Auto	167.7	10.2	6.08%	2.12%
Personal loan	112.1	16.4	14.65%	5.82%
Credit card	71.6	10.1	14.10%	6.82%
Overdraft	29.7	5.1	17.25%	8.73%

Table 5: ECL by product. Mortgages carry the largest absolute ECL due to volume; overdrafts carry the highest coverage ratio due to combined high PD and high LGD.

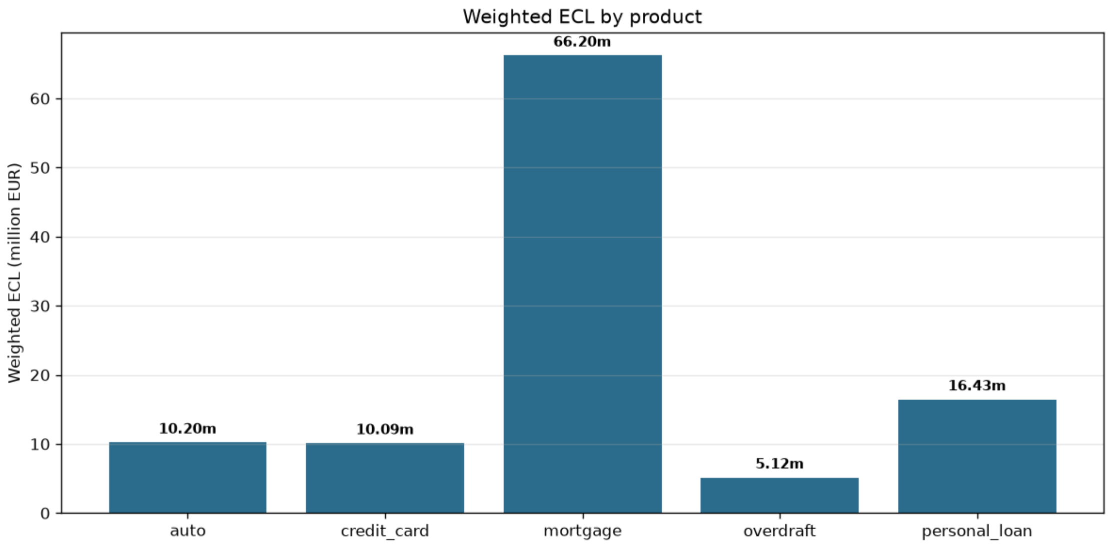

Figure 4: Weighted ECL by product. Mortgage dominates absolute provisioning; unsecured revolving products dominate coverage percentages.

4.4 Scenario sensitivity

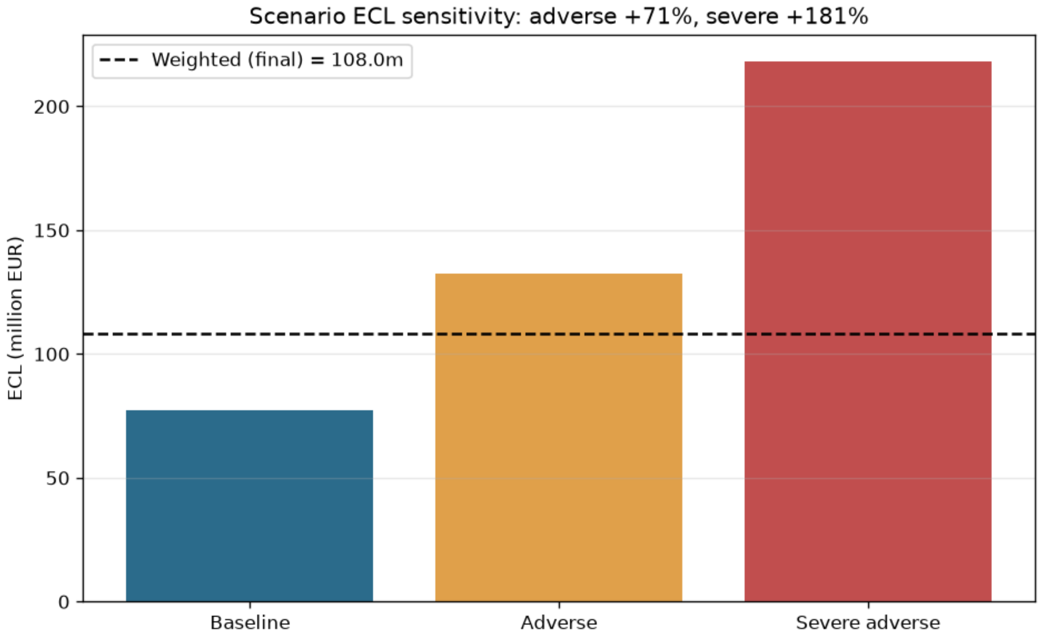

Figure 5: Portfolio ECL across the three macro paths with the weighted line overlaid. The adverse scenario roughly doubles the baseline provision; severe adverse triples it. The weighted line sits closer to the baseline because the 60%-weight baseline dominates the linear combination.

4.5 Stage-migration cliff

A recurring question in IFRS 9 is: how much would provisions jump if a stage-1 loan migrated to stage 2? For every performing stage-1 loan we compute its hypothetical stage-2 ECL (lifetime PD $\times$ LGD $\times$ EAD, lifetime-discounted) and compare against its stage-1 ECL (12-month).

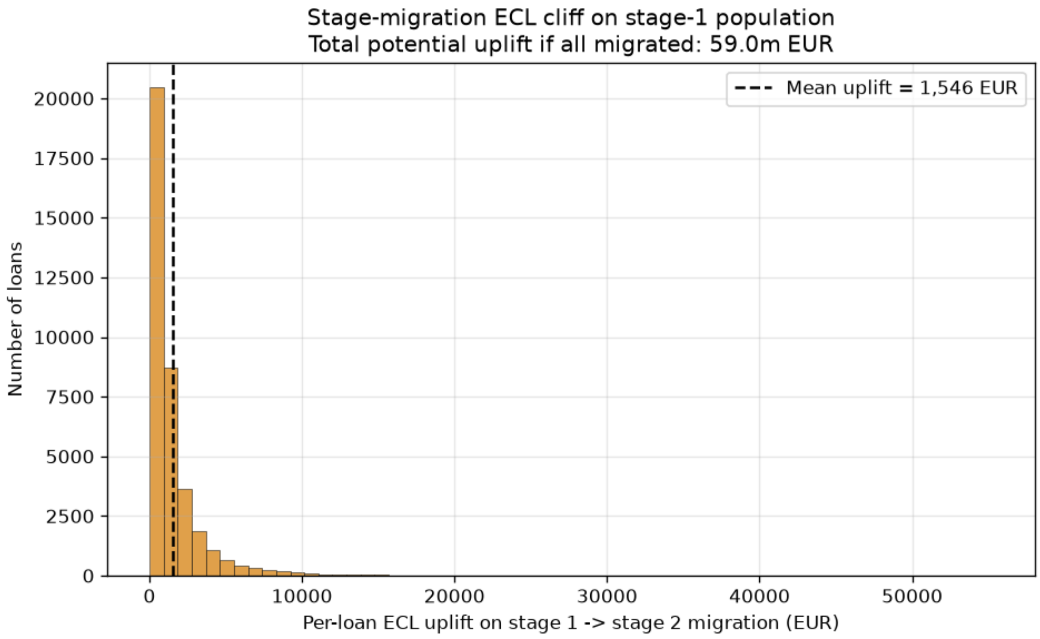

Figure 6: Distribution of per-loan ECL uplift if a stage-1 loan migrates to stage 2. Mean uplift is 1,546 EUR per loan. If all stage-1 loans migrated, total provisions would rise by 59m EUR (+55% over the current weighted total). This distribution is what sensitivity dashboards report to the CFO under IFRS 9 governance.

5 IFRS 9 fit

- **Point-in-Time (PIT) PD:** The macro-conditional PD multipliers deliver a PIT PD by scenario, which is the IFRS 9 prescription (as opposed to Through-The-Cycle PD used for Basel IRB capital).
- **Forward-looking macro:** Three scenarios with fixed weights and stress multipliers produce the probability-weighted provision required by IFRS 9.5.5.17.
- **Stage 1 (12m) vs Stage 2/3 (lifetime):** Correctly distinguished, with stage 3 assigned PD = 1.
- **Discounting to reporting date:** Applied at 3% effective rate.
- **Not fully compliant on:** the macro scenarios are deterministic multipliers; a production framework would use stochastic simulation of macro paths and rebuild the term-structure of PD conditional on each simulated path. Discount rate should be the effective interest rate at origination, not a portfolio average.

6 Limitations

- PD, LGD, and EAD are treated as independent per-loan draws. In reality all three are correlated with the same macro state, which produces higher tail ECL than the multiplicative stress applied here.
- Lifetime PD uses a constant-hazard approximation across remaining maturity. A production term-structure model would estimate age-dependent hazards.

- Behavioural revolving-limit dynamics (limit changes over time, borrower requesting increases) are not modelled.
- No collateral value re-appraisal over the lifetime horizon.
- Stage classification is snapshot-based; a lifetime measurement requires modelling stage-migration probabilities.

7 Files

- `generate_portfolio.py` — synthetic performing portfolio + macro scenarios
- `compute_ecl.py` — end-to-end ECL pipeline with staging, lifetime PD, discounting, three scenarios, weighting
- `outputs/` — loan-level ECL, portfolio-level aggregates by stage and product
- `figures/` — six diagnostic plots